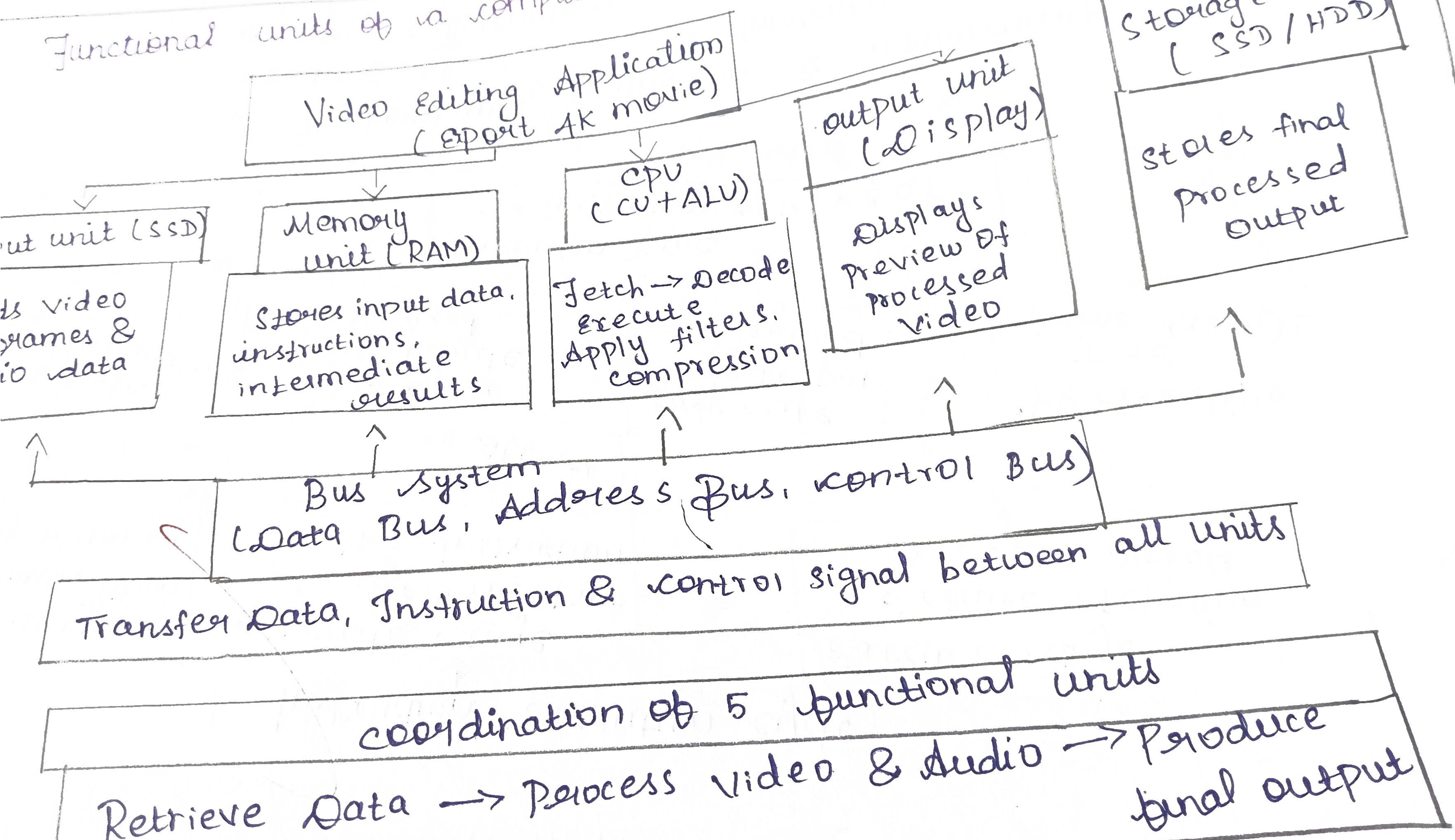
Functional units of a computer (Video Editing Application)



Instruction Execution cycle & Performance Parameters

Instruction execution cycle

Fetch stage
fetch instruction from memory

Decode stage
Decode & interpret the instruction

Execute stage
Execute operation & store result

Affects

CPI
(clock cycle per instruction)

clock cycles
Total cycle used

clock rate
Cycle per second

CPU Execution Time
(Time taken)

CPU Execution Time

$$= \frac{\text{Instruction} \times \text{CPI} \times \text{CCT}}{\text{clock rate}}$$

Performance Evaluation & Improvement

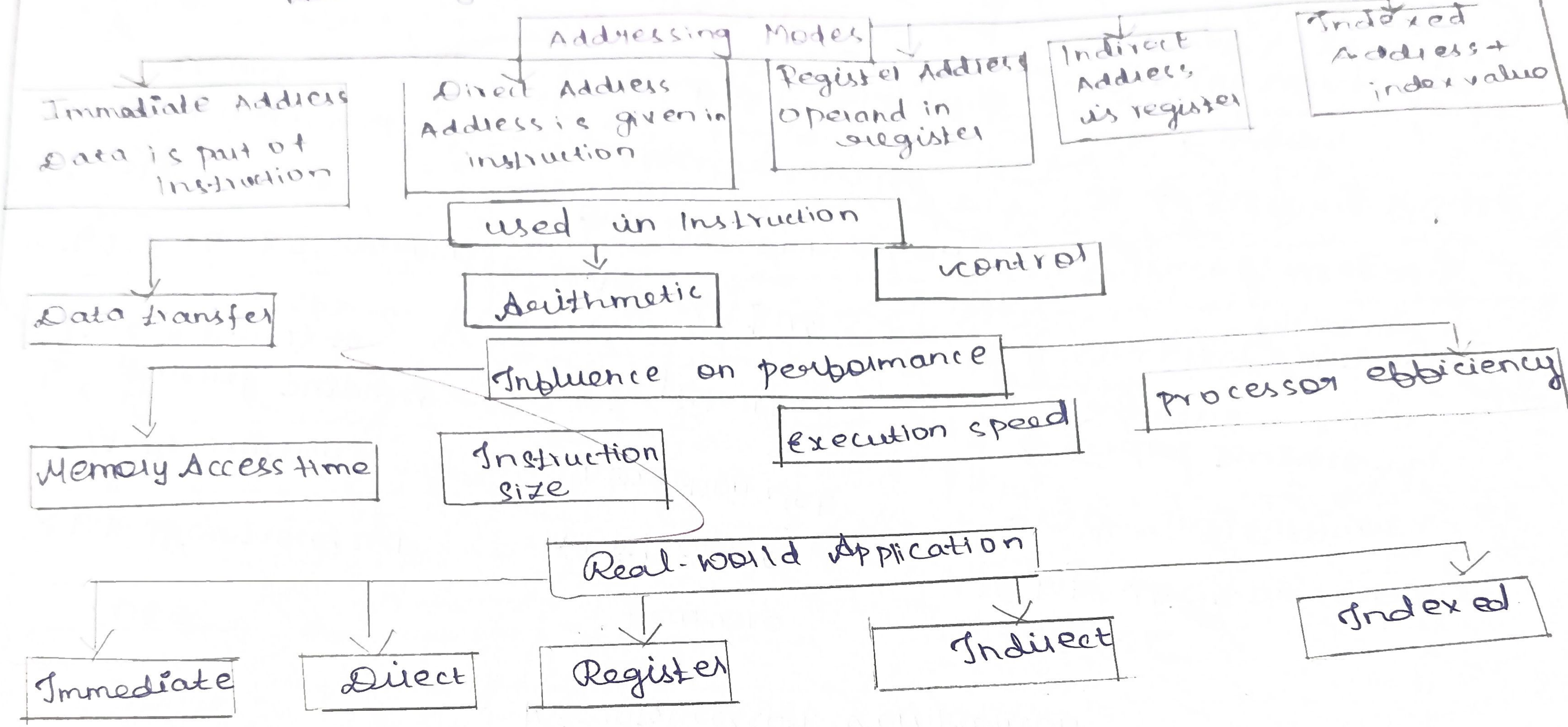
* Reduce CPI
* optimize Instruction
* Increase clock rate
* Efficient Hardware

3. Comparison of 8085 and 8086

Microprocessor comparison

8085	Feature	8086
8 bit Microprocessor	Architecture	16-bit Microprocessor
74 Instructions	Instruction set	~130 Instructions
6 General purposes (B, C, D, E, H, L) +2 (PC, SP)	Registers	8 General purpose + 4 Segment
16-bit (64 KB Memory)	Address Bus	20-bit (1 MB Addressable Memory)
Simple Memory	Memory organization	Segment Memory
5 Addressing Modes	Addressing Modes	12 Addressing Modes
Slower	Speed & Performance	Faster
Simple	Application	Multitasking & complex Real-time Application

Addressing Modes and their Applications



5. Binary and Hexadecimal Number system in computer system

Number systems

Equivalent Representation

Binary Number System

Used for Instruction Representation

Stored in Memory

CPU Instruction Interpretation

Why Hexadecimal is preferred

Compact Representation
Easy to Read and write

Hexadecimal Number System

Human friendly Representation of Binary

Data conversion

Easier Debugging and Analysis

Supports

Instruction Representation

Memory organization

CPU Execution

Debugging & Testin